

12 (**long**) An air-filled hollow conducting sphere has outer and inner radii of $R_1 = 5.0$ cm and $R_2 = 4.0$ cm with its outer surface connected to earth, as shown in Fig. 7. A conducting concentric sphere with a radius of $R_3 = 2.0$ cm is placed inside the hollow sphere and it has a positive charge of $Q = 3.0 \times 10^{-8}$ C.

- (a) What is the electric field at $R = 1.0$ cm, $R = 3.0$ cm and $R = 6.0$ cm? [15]
- (b) What is the capacitance of the concentric spheres? [7]
- (c) What is the electrostatic energy stored in the concentric spheres? [8]

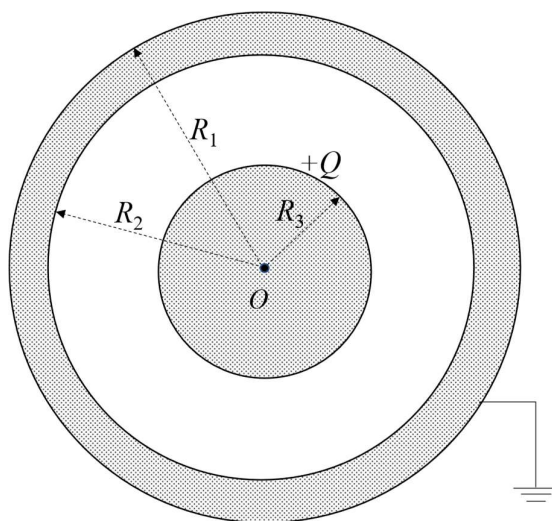


Fig. 7

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